

Modality 4: Voice

Brief: Voice Activated Ship Interior

Description:

For this week's example, we are going to build a rudimentary voice assistant for our spaceship. Simply, say the magic word and turn on a light in a spaceship with your voice.

Resources:

OpenAI Whisper:

<https://github.com/openai/whisper>

How AI is Changing Language:

<https://www.theguardian.com/books/ng-interactive/2026/jul/04/future-of-fiction-next-great-novel-ai-language-chat-gpt>

The Acousmatic Question and the Will to Datafy:

<https://sterneworks.org/wp-content/uploads/2024/01/Sterne-Sawhney-AcousmaticQuestion.pdf>

Example 1: Voice Controlled Interior



What's Happening:

1. Whisper running via Python
2. Text data is sent via OSC

3. OSC data is sent via serial port
4. OSC data is sent to a Processing sketch via the OSCp5 library.
5. Processing sketch has two images set up: a background and an overlay mask.
6. The overlay mask is activated by the command “system on”
7. The overlay mask is activated by the command “system off”

Python Instructions:

1. Make sure python and the relevant libraries are installed.
2. In Terminal, navigate to the folder where your VOICE folder is installed by using the commands:

“ls” to list folders in the directory

“cd foldername” to navigate to folders within the directory

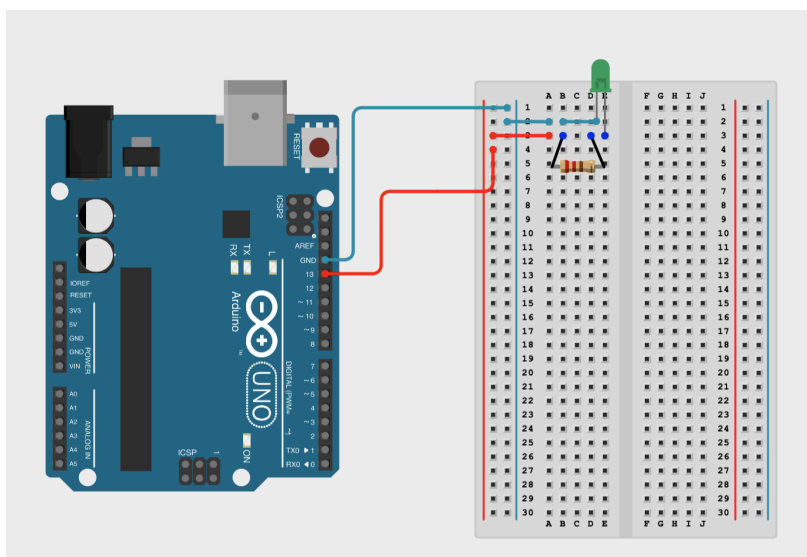
“cd ..” to return to the parent directory

3. Once you are in the folder, enter:
Python3 or pip3 voice.py

Processing Instructions:

1. Open the Processing software
2. Open the example file VOICE_E1.pde
3. Press the play button, with the python code running.
4. If all goes well, you should be able to turn on the lights of the cockpit with the commands “system on” and “system off”.

Example 2: Voice Controlled Interior w/ Hardware



Arduino Connections:

1. With the Arduino unpowered, connect the GND on the digital pin side of the arduino to the ground strip on the breadboard.
2. Connect this to the short leg of a green LED.
3. Connect pin D13 to a 220ohm resistor, then to the long leg of the green LED.

Arduino Code:

1. On your computer, open the Arduino IDE
2. Plug the Arduino in via USB
3. Open the example file Voice_AR.ino

Python Instructions:

1. Make sure python and the relevant libraries are installed.
2. In Terminal, navigate to the folder where your VOICE folder is installed by using the commands:

“ls” to list folders in the directory

“cd foldername” to navigate to folders within the directory

“cd ..” to return to the parent directory

3. Once you are in the folder, enter:
Python3 or pip3 voice.py

Processing Instructions:

1. Open the Processing software
2. Open the example file VOICE_E2.pde
3. Press the play button, with the python code running.
4. If all goes well, you should be able to turn on the lights of the cockpit on screen and the green LED on with the commands “system on” and “system off”.

To Think About:

1. Whose voices work best with Voice Recognition and don't? Are there certain accents or expressions registered incorrectly? How can you know based on the technology?
2. Try speaking at the same time as a group - notice the python transcription. Does it capture everyone's voices?
3. Where might you want to apply voice recognition as an interface? How do you feel if it's always listening?
4. Often, voice models tend to sound more human than not. Is this better or worse for society overall?

Challenges:

1. Go back to the first example code. Can you modify it such that the commands “Next” and “Previous” scroll through the images?
2. Instead of using the FSR, can you use the command “Zoom In” and “Zoom out” to control the scale of the image?
3. How can you use the capacitive touch sensor to activate and deactivate listening mode?